

EE3621: Digital Signal Processing

Lecture 20: Lattice & Lattice-Ladder Structures & Finite Word-Length Effects (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Convert** FIR direct form coefficients to lattice reflection coefficients k_m using the step-down algorithm.
 2. **Evaluate** IIR filter stability directly from reflection coefficients.
 3. **Construct** signal flow graphs for FIR lattice and IIR lattice-ladder filters.
 4. **Quantify** round-off noise power and identify limit cycle conditions.
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2. MATHEMATICAL FOUNDATIONS

2.1 FIR Lattice Recursion & Step-Down Algorithm

- **Lattice Recursion:**

$$\begin{aligned}A_m(z) &= A_{m-1}(z) + k_m z^{-1} B_{m-1}(z) \\ B_m(z) &= z^{-m} A_m(z^{-1}) \quad (\text{Reverse Polynomial})\end{aligned}$$

- **Step-Down Algorithm (Direct $\rightarrow$ Lattice):**

Given $A_N(z) = 1 + \sum_{i=1}^N \alpha_N(i) z^{-i}$: 1. Set $k_N = \alpha_N(N)$. 2. For $m = N, N-1, \dots, 2$:

$$\alpha_{m-1}(i) = \frac{\alpha_m(i) - k_m \alpha_m(m-i)}{1 - k_m^2}, \quad i = 1, 2, \dots, m-1$$

$$k_{m-1} = \alpha_{m-1}(m-1)$$

2.2 Finite Word-Length Effects Summary

1. **Coefficient Quantization:** Shifts poles in the z -plane; high-order direct forms are severely sensitive; lattice and cascade forms are highly robust.
2. **Round-off Noise:** Fixed-point truncation error modeled as white noise with variance $\sigma_e^2 = \frac{2^{-2B}}{12}$ for B -bit quantization. Output noise power:

$$\sigma_y^2 = \sigma_e^2 \sum_{n=0}^{\infty} |h[n]|^2 = \sigma_e^2 \frac{1}{2\pi j} \oint H(z)H(z^{-1})z^{-1}dz$$

1. **Limit Cycle Oscillations:** Nonlinear oscillations in recursive filters caused by arithmetic quantization (deadbands) and two's complement overflow. Mitigated by using saturation arithmetic.

Pictorial Representations

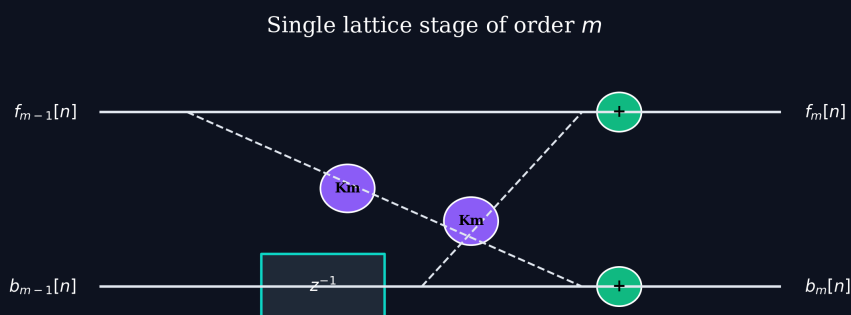


Figure 1: Fir Lattice Stage

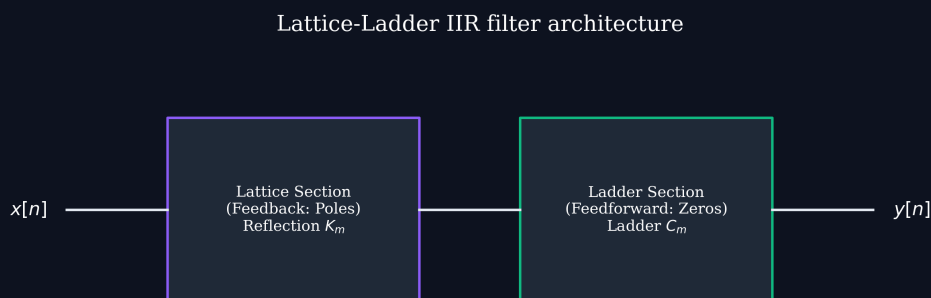


Figure 2: Lattice Ladder

3. WORKED NUMERICAL EXAMPLES

Example 20.1: Step-Down Conversion to Reflection Coefficients

Problem: Find the reflection coefficients k_1, k_2, k_3 for the FIR filter:

$$A_3(z) = 1 - 0.9z^{-1} + 0.64z^{-2} - 0.5z^{-3}$$

Solution:

• **Stage 3:**

$$k_3 = \alpha_3(3) = -0.5. \quad 1 - k_3^2 = 1 - (-0.5)^2 = 1 - 0.25 = 0.75.$$

• **Compute $\alpha_2(i)$ for $m = 3$:**

$$\begin{aligned} * \alpha_2(1) &= \frac{\alpha_3(1) - k_3 \alpha_3(2)}{0.75} = \frac{-0.9 - (-0.5)(0.64)}{0.75} = \frac{-0.9 + 0.32}{0.75} = \frac{-0.58}{0.75} = -0.7733. \\ * \alpha_2(2) &= \frac{\alpha_3(2) - k_3 \alpha_3(1)}{0.75} = \frac{0.64 - (-0.5)(-0.9)}{0.75} = \frac{0.64 - 0.45}{0.75} = \frac{0.19}{0.75} = 0.2533. \end{aligned}$$

• **Stage 2:**

$$1 - k_2^2 = 1 - (0.2533)^2 = 1 - 0.0642 = 0.9358.$$

• **Compute $\alpha_1(i)$ for $m = 2$:**

$$* \alpha_1(1) = \frac{\alpha_2(1) - k_2 \alpha_2(1)}{0.9358} = \frac{-0.7733(1 - 0.2533)}{0.9358} = \frac{-0.7733 \times 0.7467}{0.9358} = -0.6170. \quad k_1 = \alpha_1(1) = -0.6170.$$

• **Stability Check:**

$|k_1| = 0.6170 < 1$, $|k_2| = 0.2533 < 1$, $|k_3| = 0.5000 < 1$. All $|k_m| < 1 \implies$ The corresponding all-pole filter is **Stable**.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State the Schur-Cohn stability criterion for digital filters. Using the step-down algorithm, determine if the system is stable:

$$H(z) = \frac{1}{1 - 1.2z^{-1} + 0.8z^{-2} - 0.2z^{-3}}$$

(9 Marks) (b) Discuss limit cycle oscillations in recursive digital filters. Explain the difference between zero-input granular limit cycles and overflow oscillations. *(6 Marks)*

Model Answer & Step-by-Step Marking Rubric:

• **Part (a):**

* Statement of Schur-Cohn criterion: All $|k_m| < 1$ *(2 Marks)* * Step-down execution: $k_3 = -0.2 \implies 1 - k_3^2 = 0.96$ $\alpha_2(1) = \frac{-1.2 - (-0.2)(0.8)}{0.96} = \frac{-1.04}{0.96} = -1.0833$ $\alpha_2(2) = \frac{0.8 - (-0.2)(-1.2)}{0.96} = \frac{0.56}{0.96} = 0.5833 \implies k_2 = 0.5833$ $1 - k_2^2 = 1 - 0.3403 = 0.6597$ $\alpha_1(1) = \frac{-1.0833 - (0.5833)(-1.0833)}{0.6597} = \frac{-1.0833(0.4167)}{0.6597} = -0.6843 \implies k_1 = -0.6843$ *(5 Marks)* * Since $|k_1| = 0.6843 < 1$, $|k_2| = 0.5833 < 1$, $|k_3| = 0.2 < 1 \implies$ **Stable** *(2 Marks)*

• **Part (b):**

* Explanation of granular limit cycles due to quantization rounding *(3 Marks)* * Explanation of overflow oscillations and mitigation via saturation arithmetic *(3 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np
```

```
\title{\textbf{EE3621: Digital Signal Processing}} \Large Verify step-down conversion \\ \n  
a = np.array([1.0, -1.2, 0.8, -0.2])  
roots = np.roots(a)  
print("Pole Magnitudes:", np.abs(roots))  
print("All inside unit circle:", np.all(np.abs(roots) < 1.0))
```